Revised Experiments for PatternBoost Geometric Extremal Search

Experiment design note

July 4, 2026

Abstract

This note replaces the earlier broad ablation schedule for the PatternBoost geometric extremal-search project. The main observation from comparing the new multi-level experiments with the earlier code is that better bounds came from certificate-shaped search spaces, not from running a generic coordinate mutation pipeline longer. The revised experiment plan therefore uses problem-specific representations and verifiers for the three current targets: MISR LP gaps, unit-square line-stabbing gaps, and recursive guillotine hardness. Epsilon-net and representability-separation experiments are treated as separate exploratory appendix tasks, not as rows in the main three-problem table.

1 Revision principle

The previous design treated the experiment as a large uniform matrix over representations, local moves, and surrogate scores. That matrix was useful for debugging the harness, but it is not the right primary experiment for obtaining meaningful mathematical bounds. The comparison with the earlier code shows the following pattern.

1. Unit-square progress came from a one-dimensional staircase certificate that was lifted to a two-dimensional square-stabbing instance.
2. MISR progress came from changing the representation: endpoint-order sequence pairs found stronger small LP gaps than direct coordinate mutation.
3. Guillotine progress should be formulated as a threshold obstruction problem over large saved subsets, with nonseparable cores and witness-breaking moves, rather than as a flat search over rectangle coordinates.

Thus the new experimental question is not "Which cell in a generic grid wins?" The new question is:

Which certificate family, verifier, and search operator produces the strongest auditable bound?

The revised experiments should report record-quality certificates and diagnostic ablations of the specific mathematical mechanisms that produced them. Averages over many nominally identical random launches are not needed for the manuscript.

2 Global requirements

Every reported result must satisfy the following requirements.

1. The reported value is recomputed by an exact verifier independent of the training loop.
2. The certificate is standalone: it contains the geometry or combinatorial template, objective components, solver status, and a canonical hash.
3. The paper distinguishes between search score and proof score. Surrogate values, sampled checks, and neural selection scores are never presented as certified bounds.
4. Each problem has a known-baseline reproduction step before any new search is reported.
5. The generic multi-level runs may be used as diagnostics, but not as the main evidence for record bounds.

3 Experiment schedule

Phase	Output	Purpose
Artifact audit	Verified reproduction of known certificates	Confirm that the old strong constructions and the new verifier agree.
Targeted search	Candidate families and near-miss logs	Explore the certificate-shaped search spaces described below.
Mechanism ablation	Small tables comparing components inside one problem	Test which mathematical ingredient matters: representation, verifier, local move, or model proposal.
Record run	Best certificate per problem, plus verifier output	Produce final manuscript claims and certificate gallery.
Independent audit	Recomputed objective values from frozen certificates	Guard against logging, solver, or search-pipeline mistakes.

The schedule is deliberately organized by mathematical target rather than by a uniform Cartesian product of implementation choices. For each problem, the primary run should be the one that searches the most structured family first. Direct coordinate mutation is retained only as a sanity check.

4 Problem 1: unit-square line stabbing

4.1 Objective

For a finite family $\mathcal{S}$ of axis-parallel unit squares, the exact score is

$$\mathrm{score}_{\mathrm{stab}}(\mathcal{S}) = \frac{\tau(\mathcal{S})}{\tau^*(\mathcal{S})}.$$

Here $\tau(\mathcal{S})$ is the minimum number of horizontal and vertical lines stabbing all squares, and $\tau^*(\mathcal{S})$ is the LP optimum of the corresponding finite set-cover relaxation over critical horizontal and vertical line positions.

4.2 Primary experiment: staircase interval certificates

The primary unit-square experiment should be the staircase certificate search from the older successful code. A candidate is a one-dimensional interval pattern with levels. A rational measure on the line gives lower bounds a_r on the mass of intervals in level r. The certificate proves that fewer than a target number of lines cannot stab the two-dimensional product instance whenever

$$a_r + a_s \geq 1 \qquad \text{for all } r + s < k.$$

The product instance includes $I \times J$ whenever the mass of I plus the mass of J is at least one. After scaling, these products become unit squares.

This experiment should first reproduce known staircase-style certificates only after they are implemented inside the allowed code path:

$$\frac{14}{9} = 1.555\ldots, \qquad \frac{20}{13} = 1.538\ldots.$$

Those values are targets, not current evidence from this repository. The discarded `square-stabbing-14-9` package should not be used as a manuscript source.

4.3 Search variants

The staircase search should vary the following ingredients.

1. Level count and target number of required lines.
2. Interval starts and common interval length.
3. Rational support points for the one-axis measure.
4. Deterministic scans over shifted-disjoint and near-staircase families.
5. PatternBoost proposals trained on high-scoring interval patterns, not on raw square coordinates.

The raw square-coordinate Axplorer environment remains a comparison baseline. It is useful for confirming that a less structured search can find small examples such as the 3/2 pattern, but it should not be the primary search for new record certificates.

4.4 Verifier and certificate

The unit-square certificate must contain:

1. the interval levels and common length;
2. the rational measure and support;
3. level piercing numbers;
4. all inequalities $a_r + a_s \geq 1$ used in the proof;
5. the generated product squares;
6. the compressed critical line universe;

7. exact LP and IP values for the finite square instance.

The verifier should recompute both the symbolic staircase proof and the finite LP/IP confirmation. The manuscript table should report the exact ratio, the number of intervals, number of product squares, LP value, IP value, and certificate hash.

5 Problem 2: MISR LP gap

5.1 Objective

For a finite family $\mathcal{R}$ of axis-parallel rectangles, the exact score is

$$\mathrm{score}_{\mathrm{MISR}}(\mathcal{R}) = \frac{\alpha^*(\mathcal{R})}{\alpha(\mathcal{R})}.$$

Here $\alpha(\mathcal{R})$ is the maximum number of pairwise-disjoint rectangles, and $\alpha^*(\mathcal{R})$ is the standard clique LP optimum for the rectangle intersection graph.

5.2 Current baseline: endpoint-order sequence pairs

The active MISR baseline is no longer the direct-coordinate search. A candidate is encoded by two double-occurrence sequences H and V, each containing every rectangle label exactly twice. Rectangle i receives its x-span from the two positions of i in H and its y-span from the two positions of i in V. The decoded rectangle family is then sent to the ordinary exact MISR verifier.

This representation is the main useful difference from the earlier successful code. It searches endpoint orders rather than raw coordinates, so swaps, block moves, reversals, label-pair swaps, motif refreshes, and depth-weighted lifting all remain in a compact geometric search space. The role of this baseline is to test whether the endpoint-order inductive bias consistently beats direct coordinate mutation.

5.3 Primary search: sequence-pair PatternBoost

The main MISR search should train and mutate in the endpoint-sequence space, while keeping the exact verifier outside the training loop as the final authority. The preferred cell is:

$$\texttt{endpoint\_sequence\_pair} \times \texttt{sequence\_pair\_pivot} \times \texttt{exact\_lp\_gap\_pressure}.$$

The exact surrogate is expensive, but for MISR it is scientifically preferable to ranking candidates by a weak graph proxy when the target is a record LP gap. Cheaper proxies remain in the matrix only as controls.

5.4 Search tracks

The MISR experiments should have three tracks.

1. **Sequence-pair track.** Use double-occurrence endpoint sequences, motif seeds, local endpoint-order moves, recombination, and lifting. Report only decoded rectangle certificates verified by the framework scorer.

2. **Direct-coordinate diagnostic.** Keep direct coordinates as a negative control. It should explain the earlier 1.25 plateau rather than serve as the main search.

3. **Structured-program follow-up.** If sequence-pair search stalls, test triangle-free or bounded-program variants as additional structured families. These should be treated as follow-up, not as the first run.

Four-port reproduction is explicitly outside the current active plan. It can be revived as a separate archival/certificate project, but the present MISR experiments should focus on endpoint-sequence search and explicit finite rectangle certificates.

5.5 Verifier and certificate

For an unweighted finite certificate, save:

1. rectangle coordinates;
2. intersection graph;
3. triangle-free status and witness if violated;
4. all maximal clique constraints, or all edge constraints in the triangle-free case;
5. LP optimum α^* and primal solution;
6. exact maximum independent set and value α;
7. solver status, runtime, and certificate hash.

For sequence-pair candidates, also save the two sequences H, V, the decoded rectangles, and the sequence mutation/lift history when available. The manuscript should distinguish the search representation from the certified geometry: the proof object is still the decoded rectangle family.

6 Problem 3: recursive guillotine hardness

6.1 Objective

For a pairwise-disjoint family $\mathcal{R}$ of axis-parallel rectangles, define $\mathrm{save}(\mathcal{R})$ as the maximum number of rectangles that can be preserved by a recursive guillotine cutting strategy. The exact destroyed-fraction score is

$$\mathrm{score}_{\mathrm{guill}}(\mathcal{R}) = 1 - \frac{\mathrm{save}(\mathcal{R})}{|\mathcal{R}|}.$$

The generic exact DP can compute this value for finalists. However, it is not the best search objective. The stronger experimental formulation is a threshold obstruction:

$$K = \left\lfloor \frac{|\mathcal{R}|}{2} \right\rfloor + 1.$$

If there is no guillotine-separable subset of size K, then every guillotine strategy destroys at least half the rectangles.

6.2 Primary experiment: threshold obstruction search

The guillotine search should use the threshold formulation as its main driver. For each subset S, build the x-projection and y-projection overlap graphs. A vertical guillotine cut avoiding all rectangles in S exists exactly when the x-projection graph induced by S is disconnected; the horizontal case is analogous. This gives a fast recursive oracle for testing whether S is guillotine-separable.

The search score is the fraction of tested K-subsets that are nonseparable. A certified pass has zero separable K-subsets. Near-misses with very few separable witnesses are valuable and should be retained for local improvement.

6.3 Local moves

Use local moves that directly attack separable witnesses.

1. If a separable witness splits along a vertical gap, stretch or move rectangles so that the corresponding x-projection components become linked.
2. If a witness splits along a horizontal gap, do the analogous move in the y-projection.
3. Mine small nonseparable cores and preserve them during mutation.
4. Assemble recursive arrangements by placing hard cores so that every large subset contains at least one nonseparable core.
5. Use long thin blockers, pinwheel motifs, brick-wall motifs, and skew-grid templates as structured starting families.

This is different from optimizing only the best first cut. First-cut hardness is a useful diagnostic, but the final claim must be about recursive separability or exact saved count.

6.4 Verifier and certificate

The guillotine certificate should have two layers.

1. **Threshold certificate.** Prove that no K-subset is guillotine-separable, either by exact enumeration or by a nonseparable-core cover certificate.
2. **Exact-score certificate.** For smaller finalists, run the full recursive DP and report save($\mathcal{R}$) and the destroyed fraction.

For final reported arrangements, save the rectangles, K, separability oracle status, number of checked subsets, any separable witness if the candidate fails, nonseparable cores, exact DP value when computed, and certificate hash.

7 Controls and comparisons

The controls should compare mathematical mechanisms, not repeated launches. Use the following comparisons.

1. **Structured versus direct representation.** Compare the primary certificate-shaped representation against direct coordinate mutation.

2. **Model proposal versus local search.** Run the same verifier and local moves with and without PatternBoost proposals.
3. **Surrogate relevance.** Compare the search surrogate to exact scores on audited candidates and report rank correlation or clear failure cases.
4. **Known-baseline reproduction.** Every problem should first reproduce the strongest available old construction or explain why no such construction exists.
5. **Near-miss analysis.** For failed guillotine and MISR candidates, report the obstruction that prevented certification: separable witness, triangle, solver timeout, large independent set, or collapsed geometry.

8 Exploratory appendix tasks

Two exploratory tasks are implemented and have one audited overnight baseline. They should be reported separately from the three main geometric objectives.

8.1 Task A: rectangle graphs versus mixed square/segment graphs

The run searched for a rectangle intersection graph with no bounded-grid representation by a mixed family of squares and horizontal/vertical segments. The audited baseline was:

- Slurm job `16501338`;
- run root `runs/explore_overnight_20260704_051618`;
- six graph-separation rows;
- all rows completed with exit code `0:0`;
- best exact score 0.0;
- best pressure/search score 1.3845054945054946 from `graph_g3_n7_motif`.

Interpretation: the current stress objective finds difficult-looking rectangle graphs, but the bounded mixed-representation verifier did not certify a separation witness. The next useful change is a stronger non-representability verifier or a smaller structured graph family with more informative unsat cores.

8.2 Task B: halfplane epsilon-net lower-bound rediscovery

The run searched for finite point sets and thresholds where every tested k-point net misses a heavy halfplane. The audited baseline was:

- six epsilon-net rows;
- all rows completed with exit code `0:0`;
- best exact value 1.4545454545454546 from `eps_n11_t4_k3`;
- `eps_n12_t5_k3` reached the 8-hour budget without a positive exact certificate.

The repository snapshot is kept at `docs/assets/exploratory_overnight_20260704_051618/`. It contains the matrix, summaries, best certificates, renderings, and a compact final audit JSON, but not the large event streams.

9 Manuscript tables

The main manuscript should contain compact tables of certified results, not large random-launch matrices.

Problem	Primary search family	Certified objective	Baseline to reproduce
Unit-square stabbing	Staircase interval certificate and product lift	τ/τ^*	independently reimplemented staircase target; current allowed baseline is 3/2
MISR	Endpoint-order sequence pairs with exact LP-gap audit	α^*/α	Endpoint-sequence smoke certificate
Guillotine	Threshold nonseparable-subset search	destroyed fraction or no separable K-subset	Current small obstruction motifs; no stronger old certificate found

For each final result, report:

1. exact ratio or threshold statement;
2. instance size;
3. objective components;
4. verifier status;
5. runtime of the verifier;
6. certificate hash;
7. path to rendering or gallery figure.

The appendix may include diagnostic tables for direct-coordinate experiments, but these should be framed as evidence that generic search is insufficient for record bounds.

10 Artifact layout

The final artifact should organize outputs by problem and certificate family. A recommended layout is:

```
artifacts/
  unit_square/
    staircase_14_9/
      certificate.json
      verify_certificate.py
      rendered_instance.pdf
    staircase_search/
      run_config.json
      candidates.jsonl
      audited_summary.csv
```

```
misr/
  four_port/
    certificate.json
    verify_four_port.py
    verification_output.txt
  triangle_free_programs/
    programs.jsonl
    best_certificates/
    audited_summary.csv
guillotine/
  threshold_search/
    best_instance.rect
    threshold_certificate.json
    separable_witnesses.jsonl
    nonseparable_cores.jsonl
  exact_dp_audit/
    dp_certificate.json
```

Each verifier should run without training logs. Training logs are useful for reproducibility, but certificate verification must not depend on them.

11 What not to run

The following experiments should not be part of the immediate manuscript plan.

1. The full generic $3 \times 3 \times 3$ matrix as the primary evidence.
2. Long direct-coordinate MISR runs whose best result remains near small cycle-like plateaus.
3. Guillotine runs scored only by first-cut hardness.
4. Mixing epsilon-net and graph-representability exploratory rows into the main three-problem table. They belong in a separate appendix unless the paper scope is explicitly expanded.
5. Any result whose exact verifier times out or whose certificate cannot be recomputed from a standalone file.

12 Summary

The revised experimental strategy is certificate-first. Unit-square stabbing should use staircase interval certificates. MISR should use endpoint-order sequence pairs as the primary search space, with direct coordinates retained only as a diagnostic baseline. Guillotine hardness should use the threshold nonseparable-subset formulation and core-cover certificates, with full DP only for finalists. Generic PatternBoost remains useful as a proposal mechanism and diagnostic baseline, but the mathematical search space must encode the proof pattern we want the system to discover.